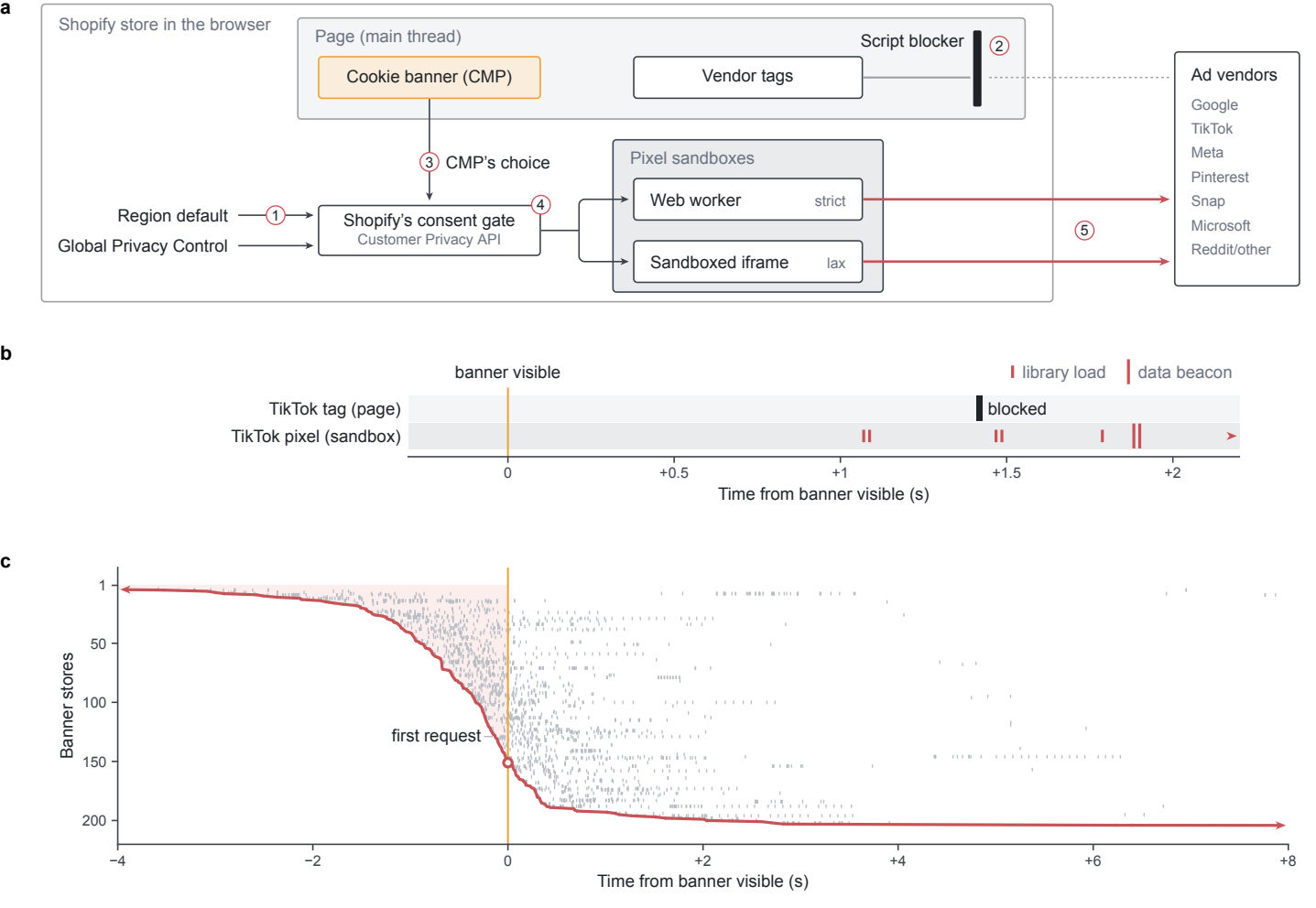
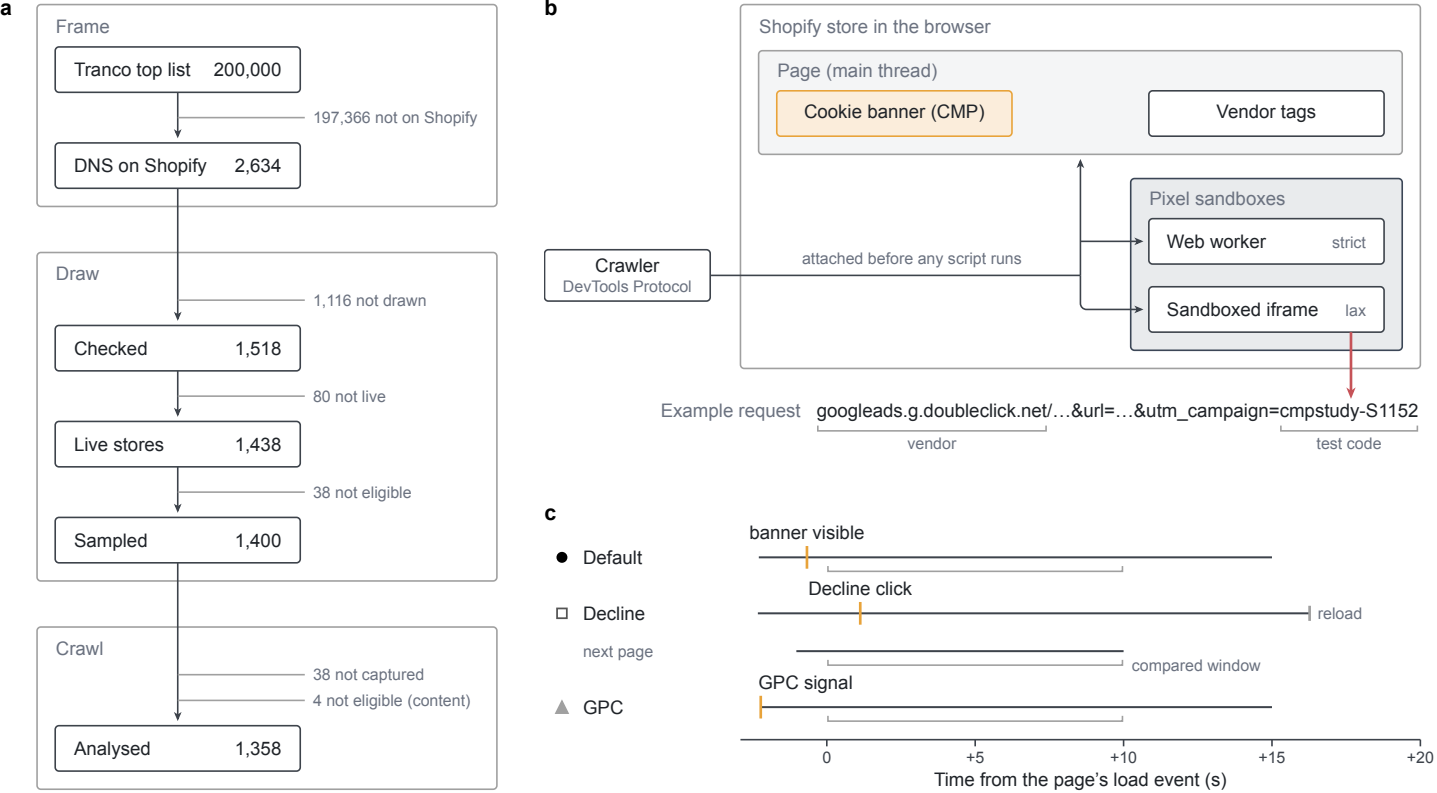
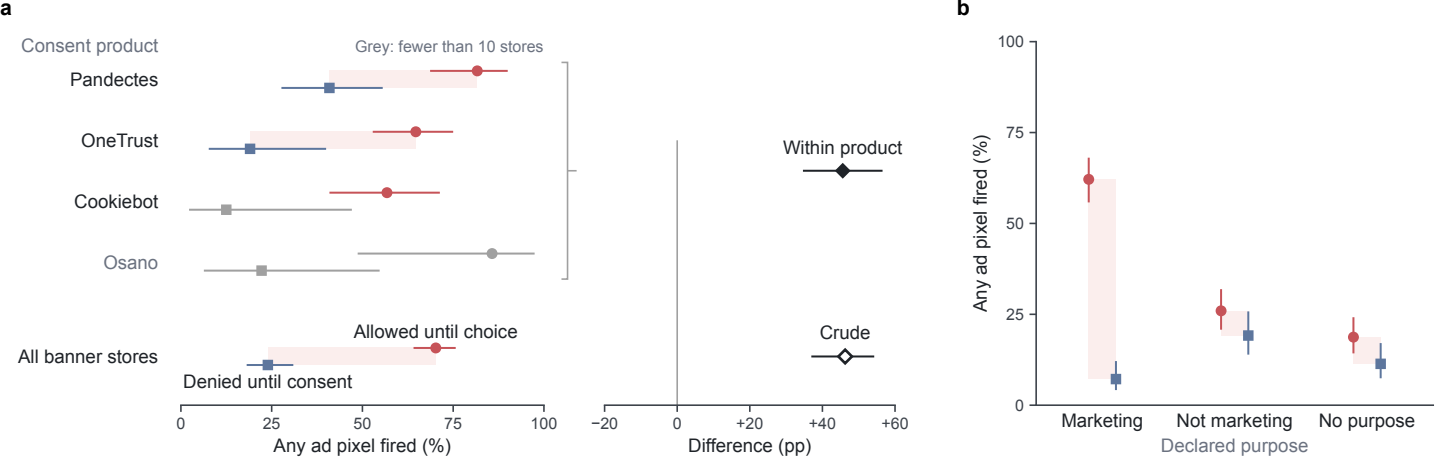
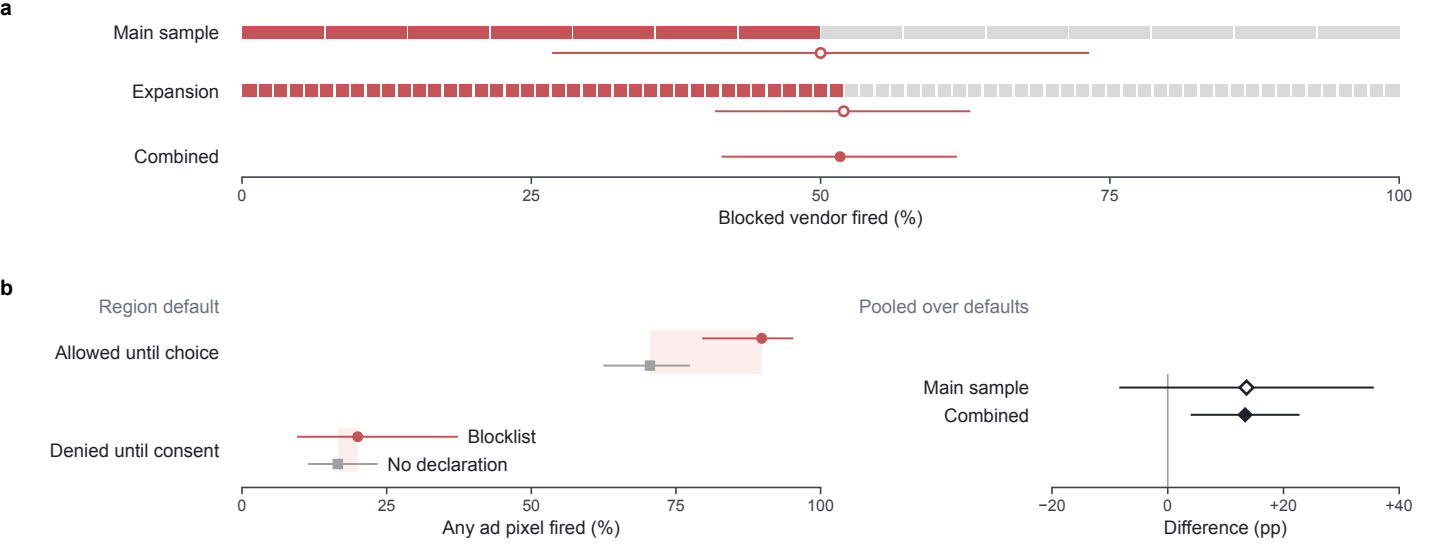
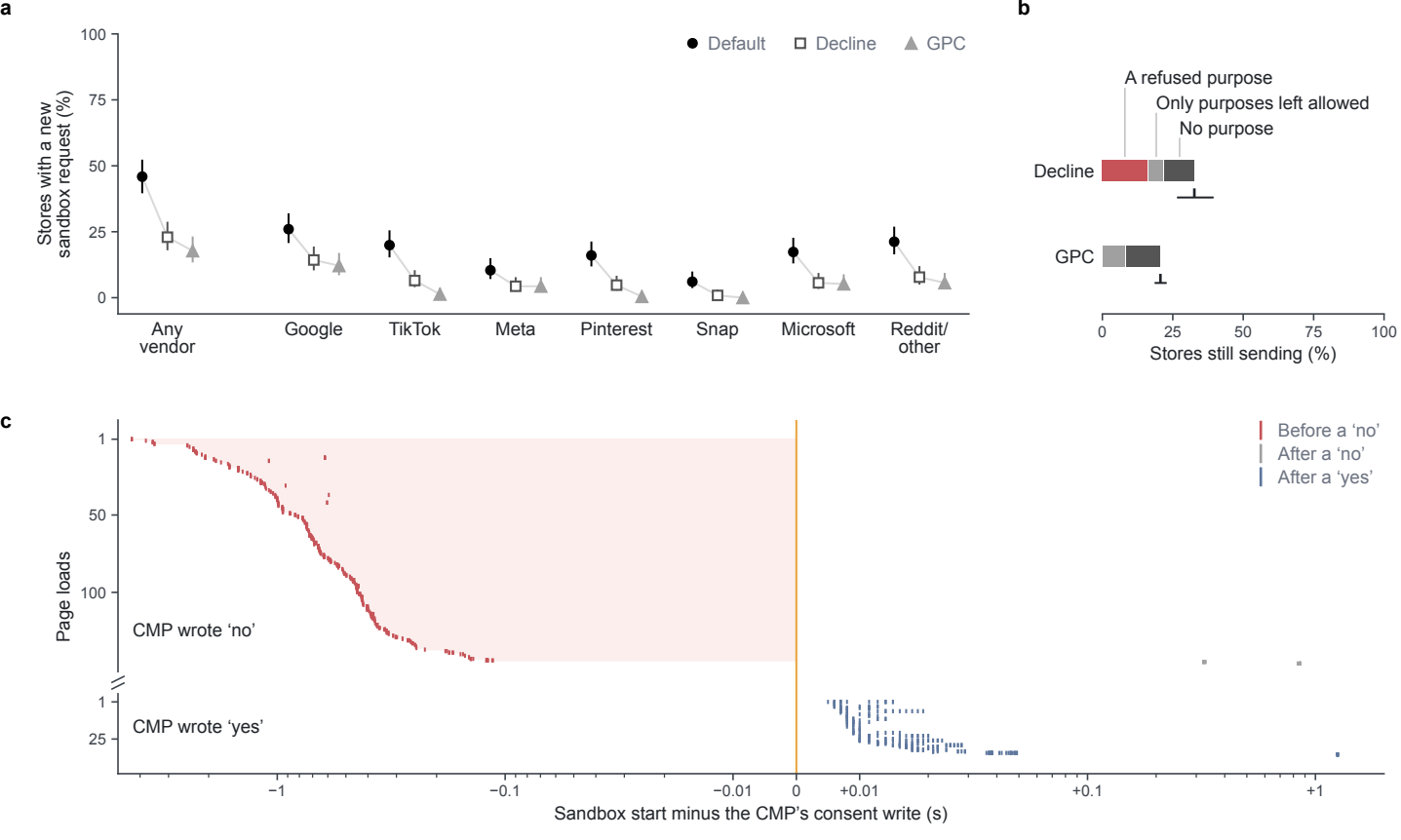
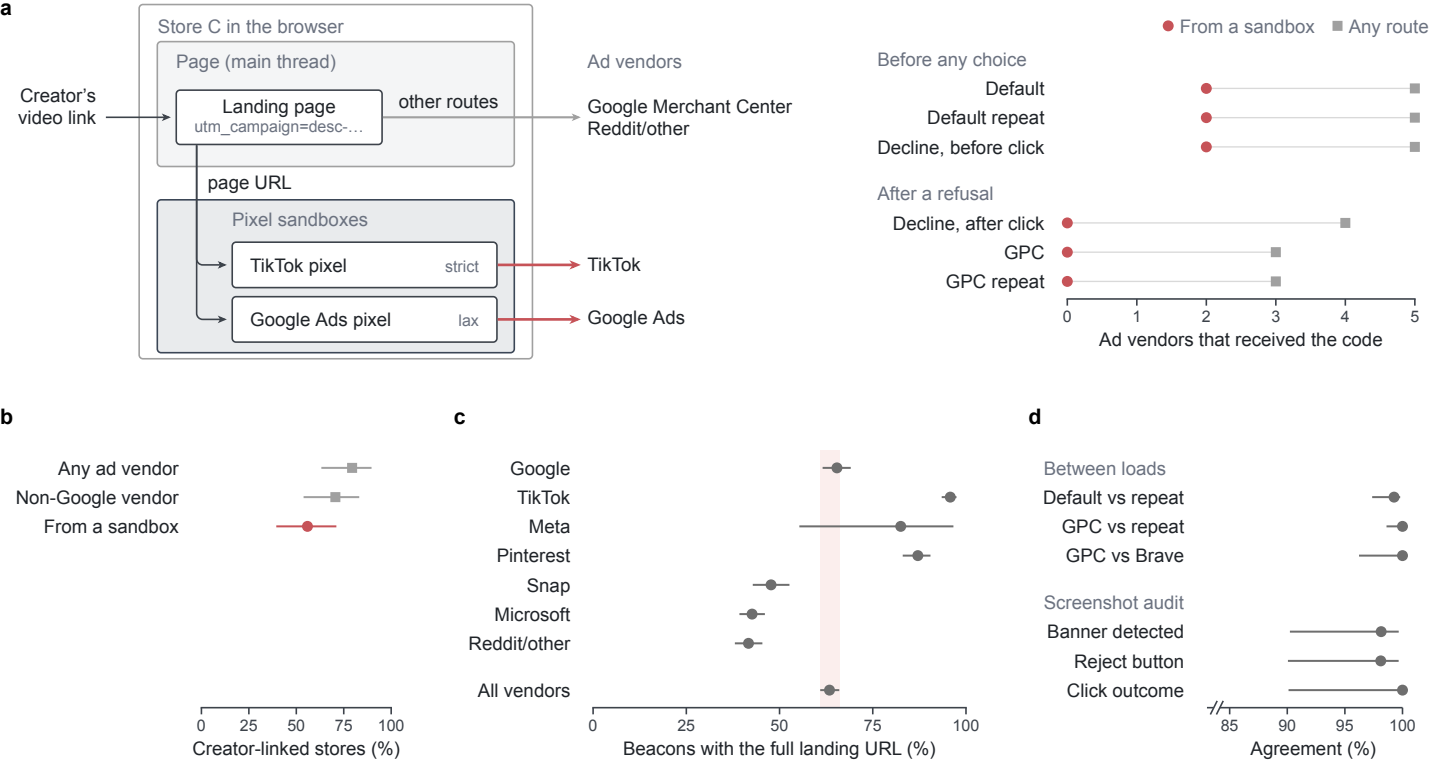

a
Shopify store in the browser
Page (main thread)
Cookie banner (CMP)
Vendor tags
Script blocker
2
Ad vendors
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
3 CMP's choice
Pixel sandboxes
Web worker strict
Sandboxed iframe lax
Region default
Global Privacy Control
1
Shopify's consent gate
Customer Privacy API
4
5
b
banner visible
library load
data beacon
TikTok tag (page)
TikTok pixel (sandbox)
blocked
0
+0.5
+1
+1.5
+2
Time from banner visible (s)
c
Banner stores
1
50
100
150
200
first request
−4
−2
0
+2
+4
+6
+8
Time from banner visible (s)

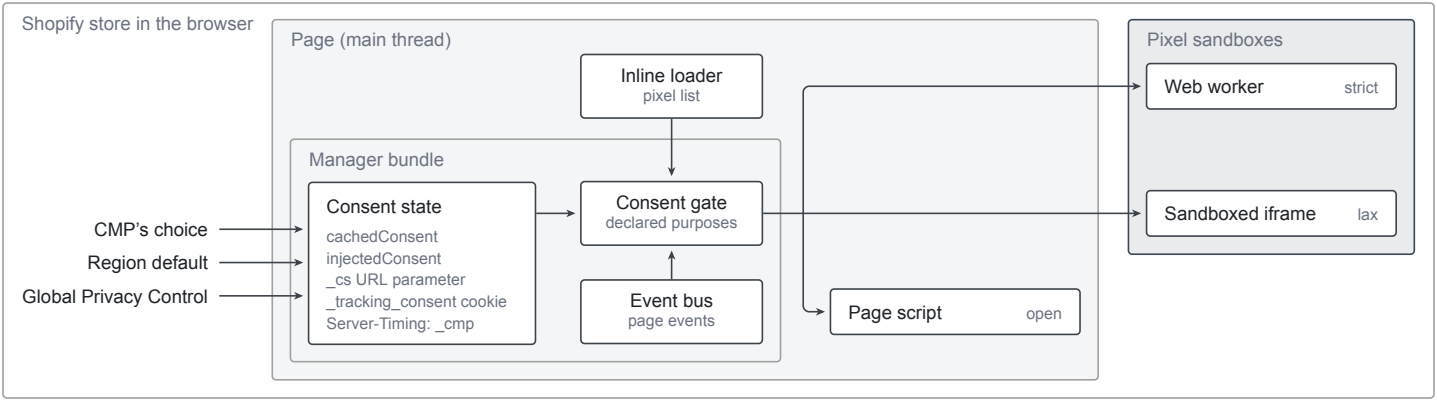

a
Frame
Tranco top list 200,000
197,366 not on Shopify
DNS on Shopify 2,634
Draw
1,116 not drawn
Checked 1,518
80 not live
Live stores 1,438
38 not eligible
Sampled 1,400
Crawl
38 not captured
4 not eligible (content)
Analysed 1,358
b
Shopify store in the browser
Page (main thread)
Cookie banner (CMP)
Vendor tags
Pixel sandboxes
Web worker strict
Sandboxed iframe lax
Crawler
DevTools Protocol
attached before any script runs
Example request googleads.g.doubleclick.net/…&url=…&utm_campaign=cmpstudy-S1152
vendor
test code
c
Default
Decline
next page
GPC
banner visible
Decline click
reload
compared window
GPC signal
0
+5
+10
+15
+20
Time from the page's load event (s)

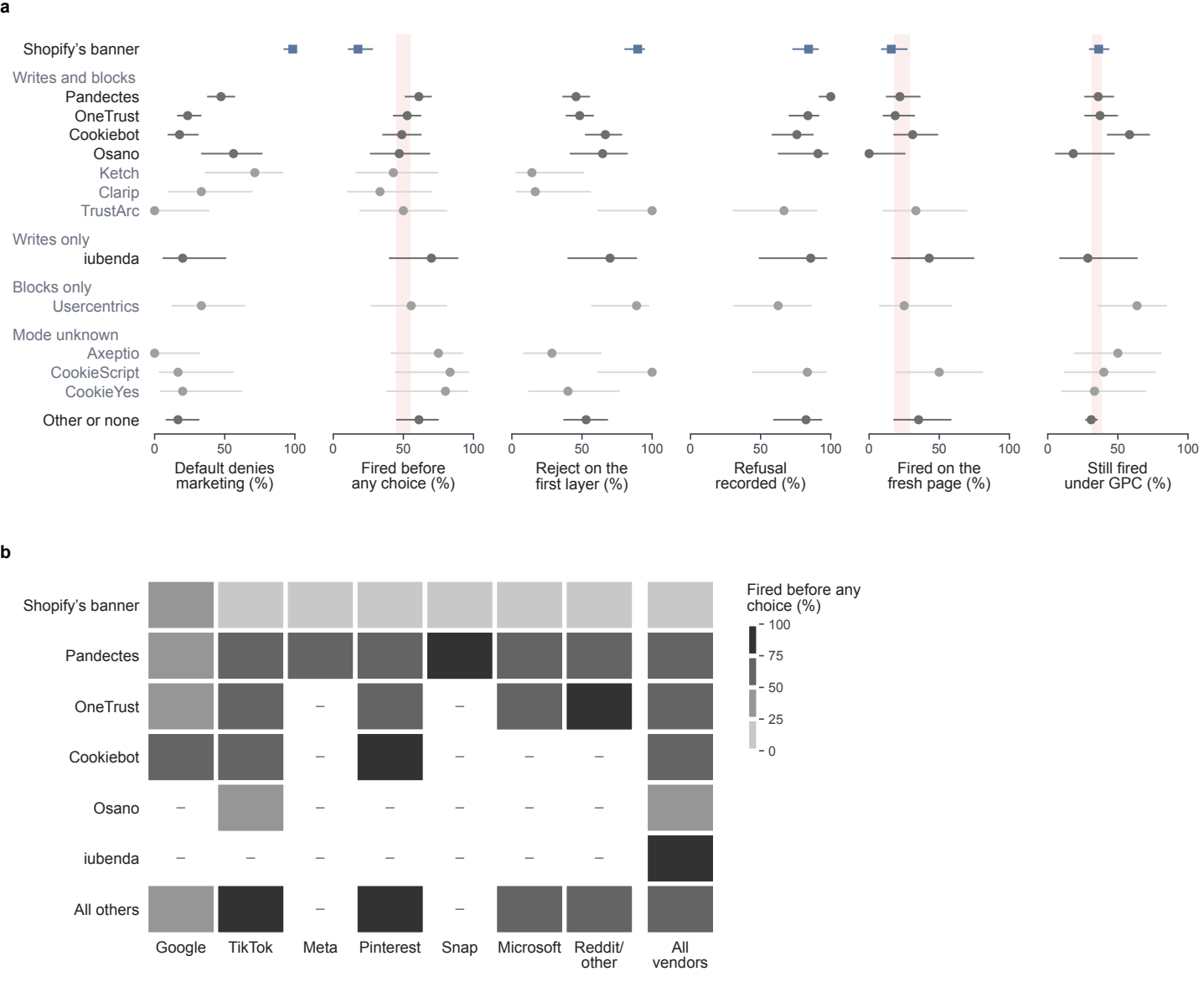

a
Consent product
Grey: fewer than 10 stores
Pandectes
OneTrust
Cookiebot
Osano
All banner stores
Allowed until choice
Denied until consent
0
25
50
75
100
Any ad pixel fired (%)
Within product
Crude
−20
0
+20
+40
+60
Difference (pp)
b
Any ad pixel fired (%)
100
75
50
25
0
Marketing
Not marketing
No purpose
Declared purpose

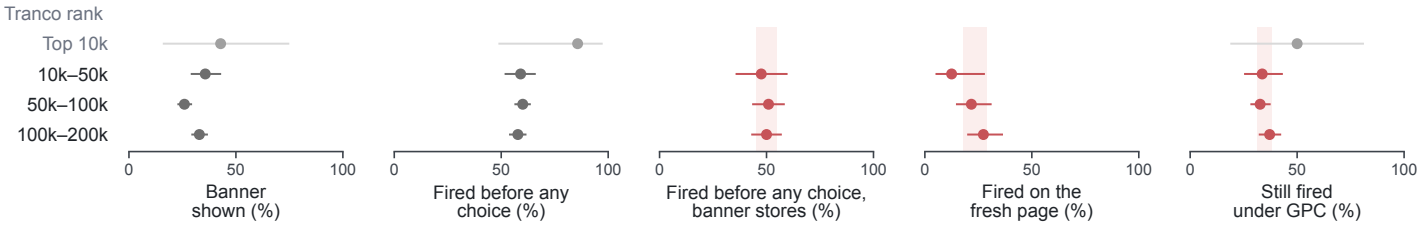

a
Main sample
Expansion
Combined
0
25
50
75
100
Blocked vendor fired (%)
b
Region default
Allowed until choice
Denied until consent
Blocklist
No declaration
0
25
50
75
100
Any ad pixel fired (%)
Pooled over defaults
Main sample
Combined
−20
0
+20
+40
Difference (pp)

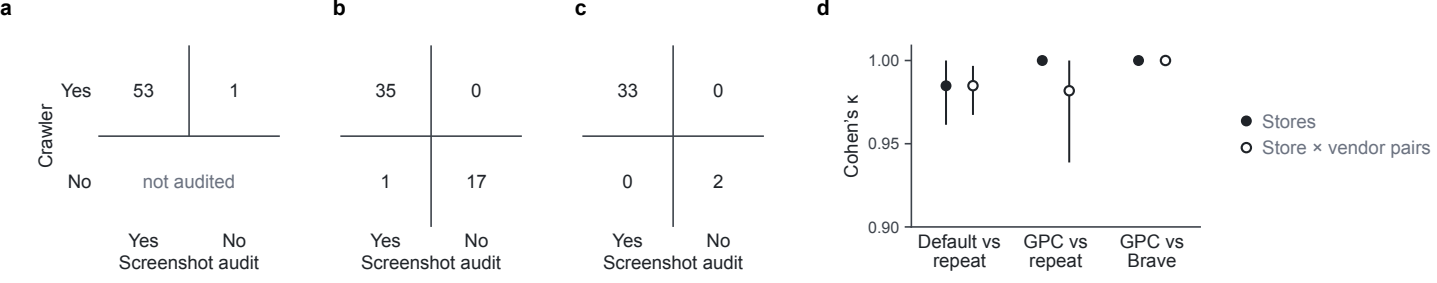

a
Default
Decline
GPC
Stores with a new sandbox request (%)
100
75
50
25
0
Any vendor
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/ other
b
A refused purpose
Only purposes left allowed
No purpose
Decline
GPC
0
25
50
75
100
Stores still sending (%)
c
Before a ‘no’
After a ‘no’
After a ‘yes’
Page loads
1
50
100
CMP wrote ‘no’
1
25
CMP wrote ‘yes’
−1
−0.1
−0.01
0
+0.01
+0.1
+1
Sandbox start minus the CMP’s consent write (s)

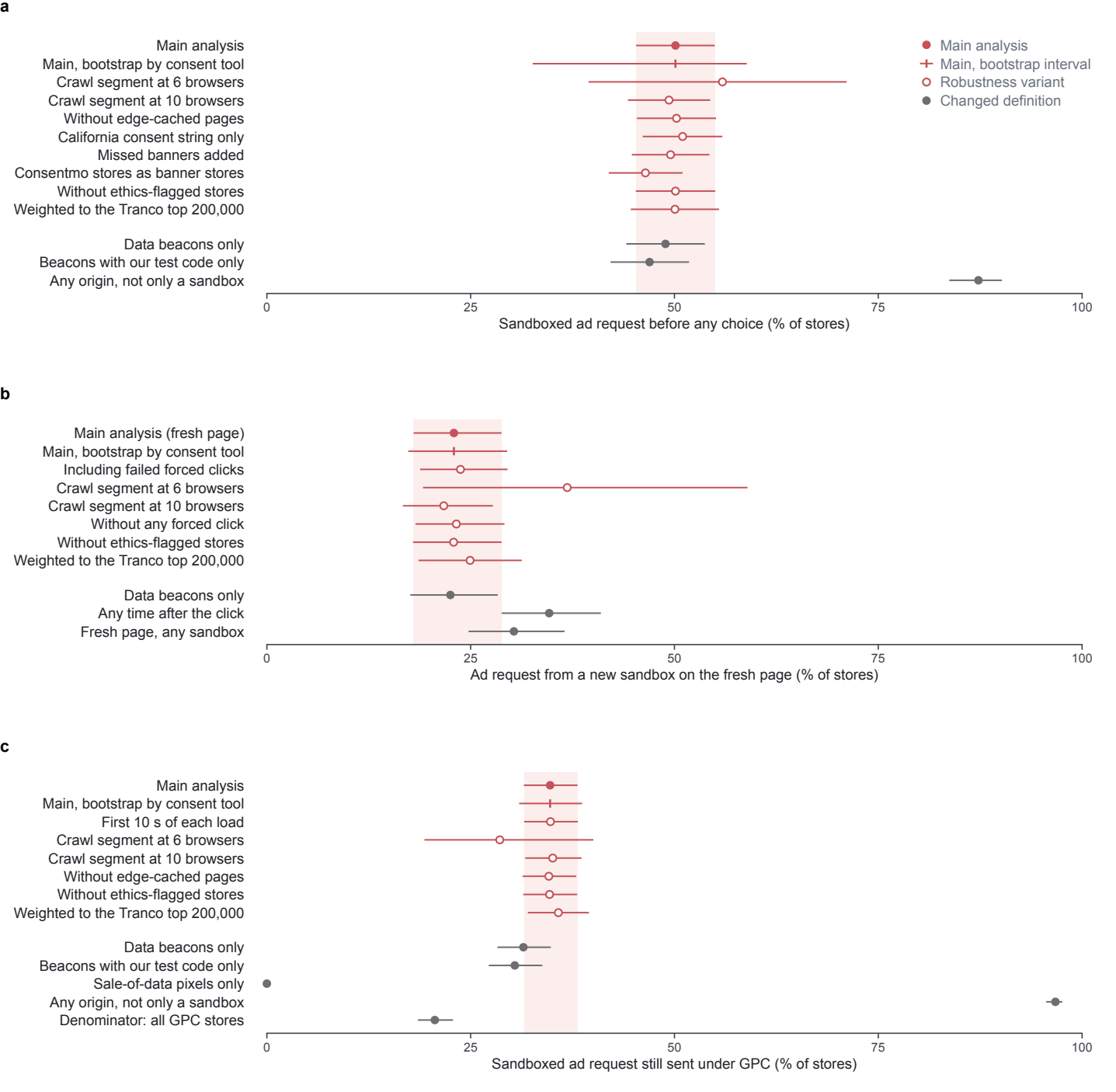

a
Store C in the browser
Page (main thread)
Creator's video link
Landing page
utm_campaign=desc-…
other routes
page URL
Pixel sandboxes
TikTok pixel
strict
Google Ads pixel
lax
Ad vendors
Google Merchant Center
Reddit/other
TikTok
Google Ads
From a sandbox
Any route
Before any choice
Default
Default repeat
Decline, before click
After a refusal
Decline, after click
GPC
GPC repeat
0
1
2
3
4
5
Ad vendors that received the code
b
Any ad vendor
Non-Google vendor
From a sandbox
0
25
50
75
100
Creator-linked stores (%)
c
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
All vendors
0
25
50
75
100
Beacons with the full landing URL (%)
d
Between loads
Default vs repeat
GPC vs repeat
GPC vs Brave
Screenshot audit
Banner detected
Reject button
Click outcome
85
90
95
100
Agreement (%)

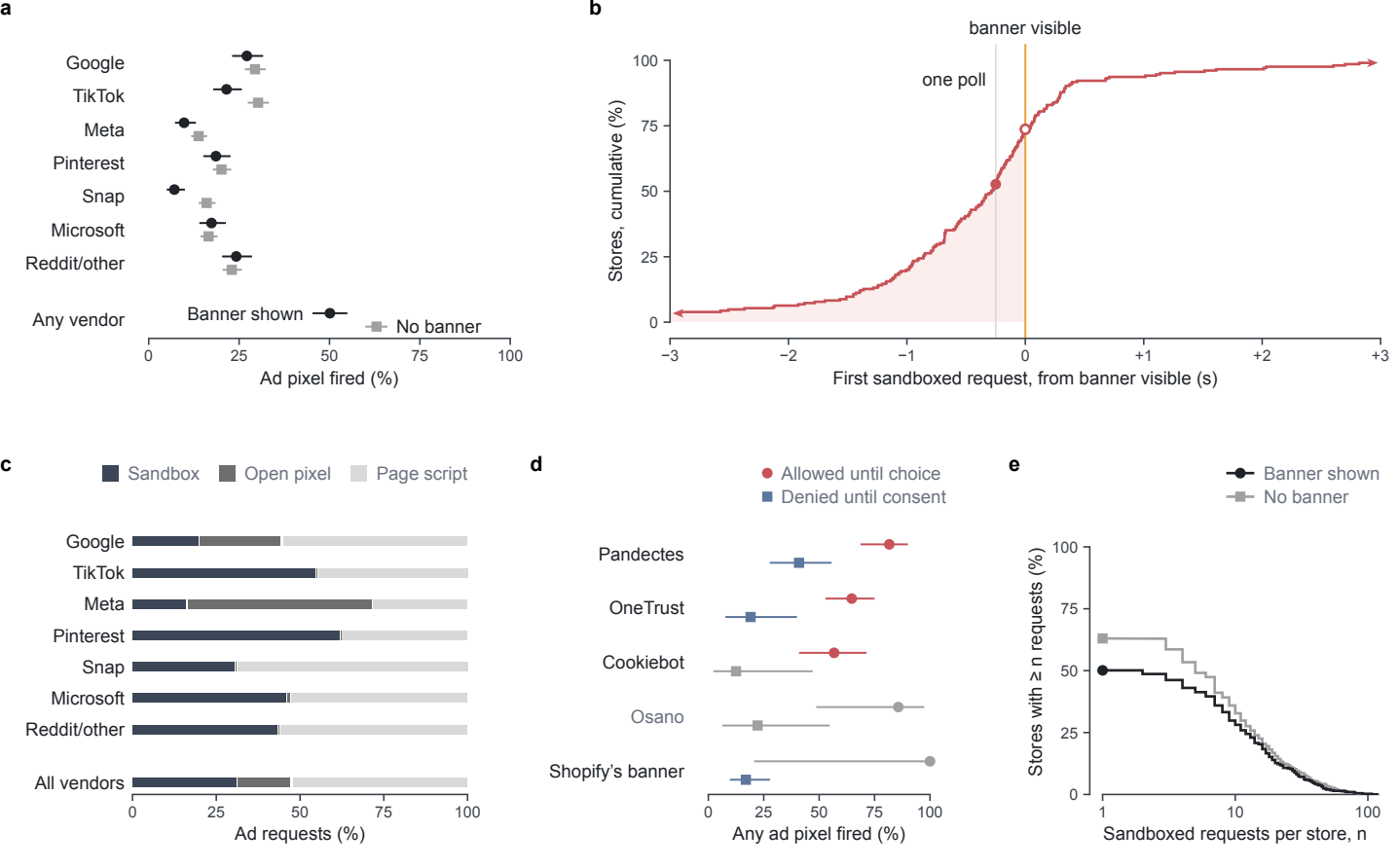

Shopify store in the browser
Page (main thread)
Inline loader
pixel list
Manager bundle
Consent state
cachedConsent
injectedConsent
_cs URL parameter
_tracking_consent cookie
Server-Timing: _cmp
Consent gate
declared purposes
Event bus
page events
CMP's choice
Region default
Global Privacy Control
Pixel sandboxes
Web worker
strict
Sandboxed iframe
lax
Page script
open

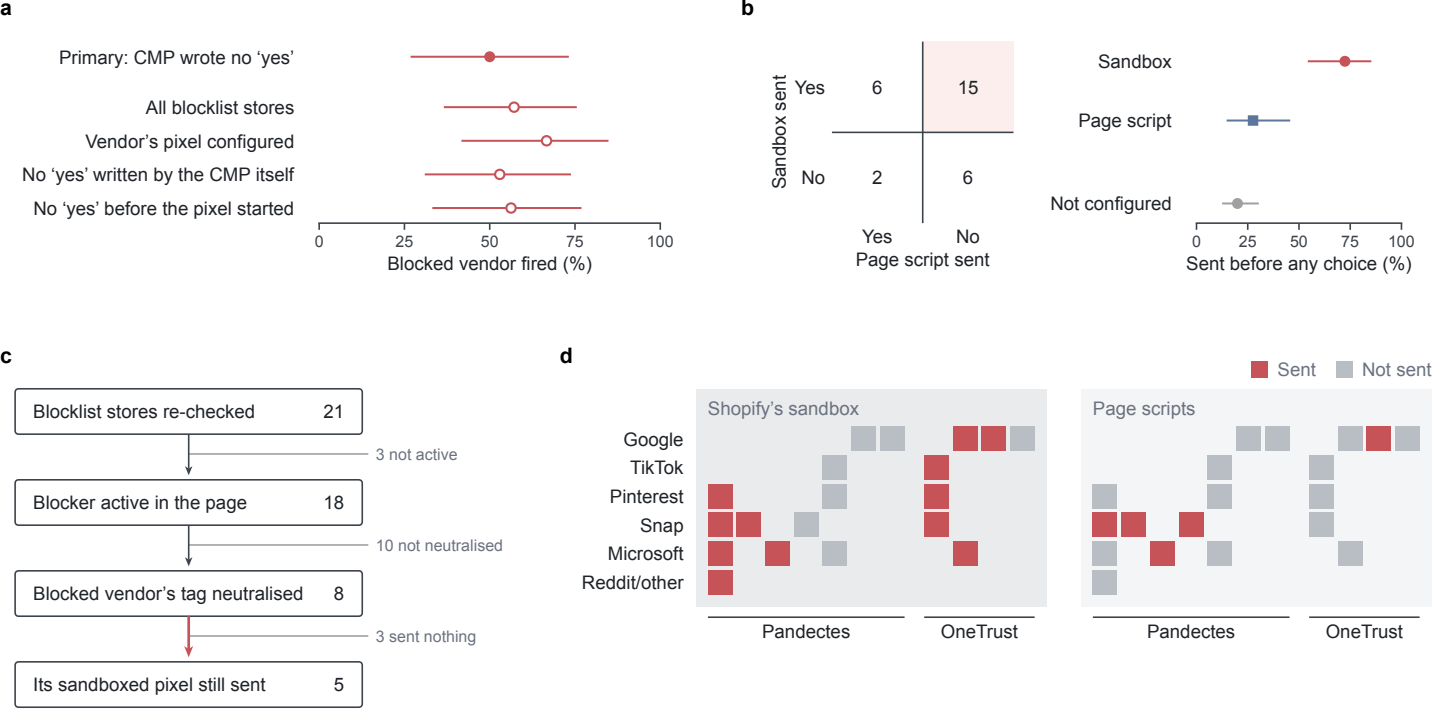

a
Shopify's banner
Writes and blocks
Pandectes
OneTrust
Cookiebot
Osano
Ketch
Clarip
TrustArc
Writes only
iubenda
Blocks only
Usercentrics
Mode unknown
Axeptio
CookieScript
CookieYes
Other or none
0
50
100
Default denies marketing (%)
Fired before any choice (%)
Reject on the first layer (%)
Refusal recorded (%)
Fired on the fresh page (%)
Still fired under GPC (%)
b
Shopify's banner
Pandectes
OneTrust
Cookiebot
Osano
iubenda
All others
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/ other
All vendors
Fired before any choice (%)
100
75
50
25
0

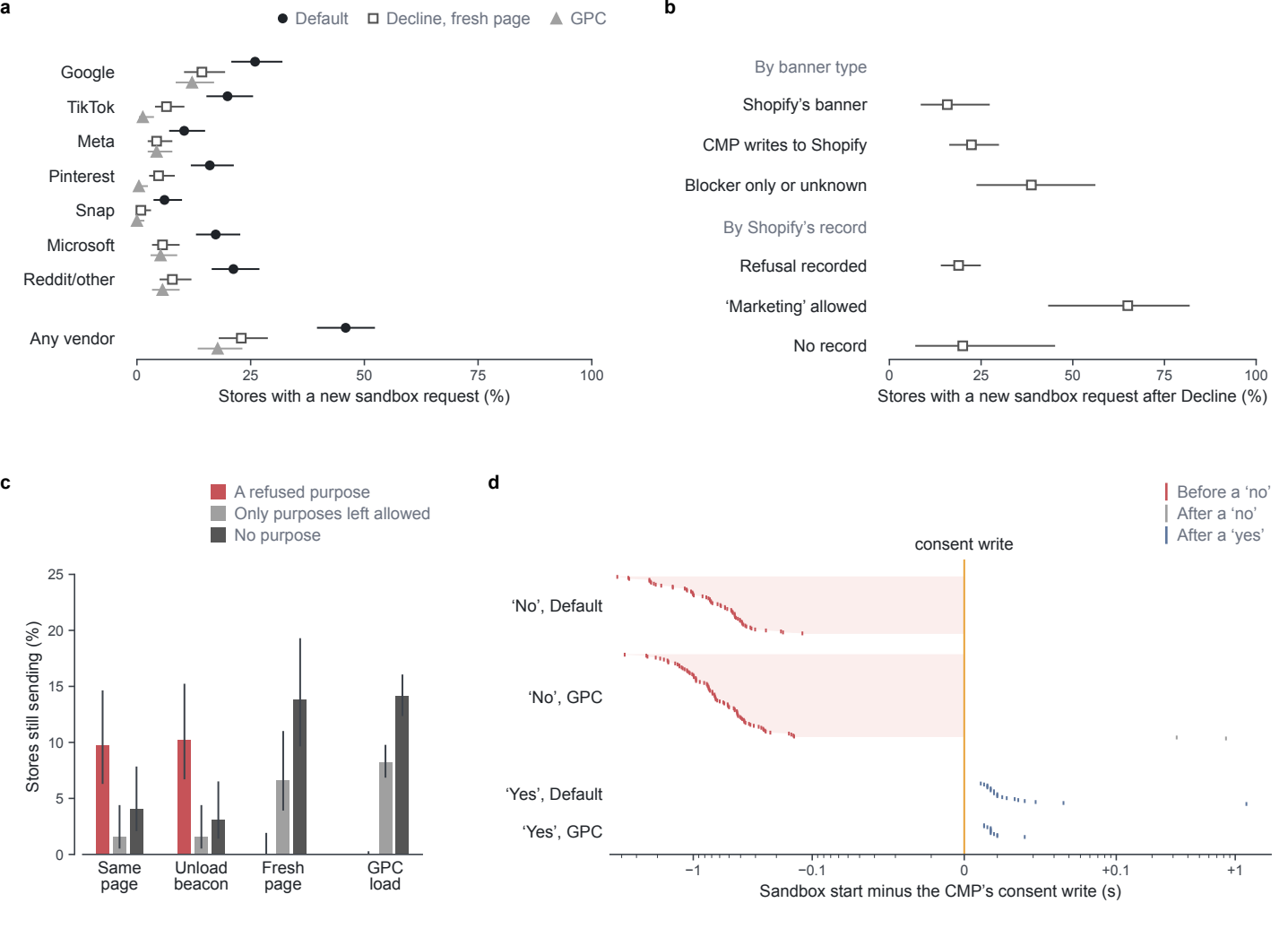

Tranco rank
Top 10k
10k–50k
50k–100k
100k–200k
0
50
100
Banner shown (%)
Fired before any choice (%)
Fired before any choice, banner stores (%)
Fired on the fresh page (%)
Still fired under GPC (%)

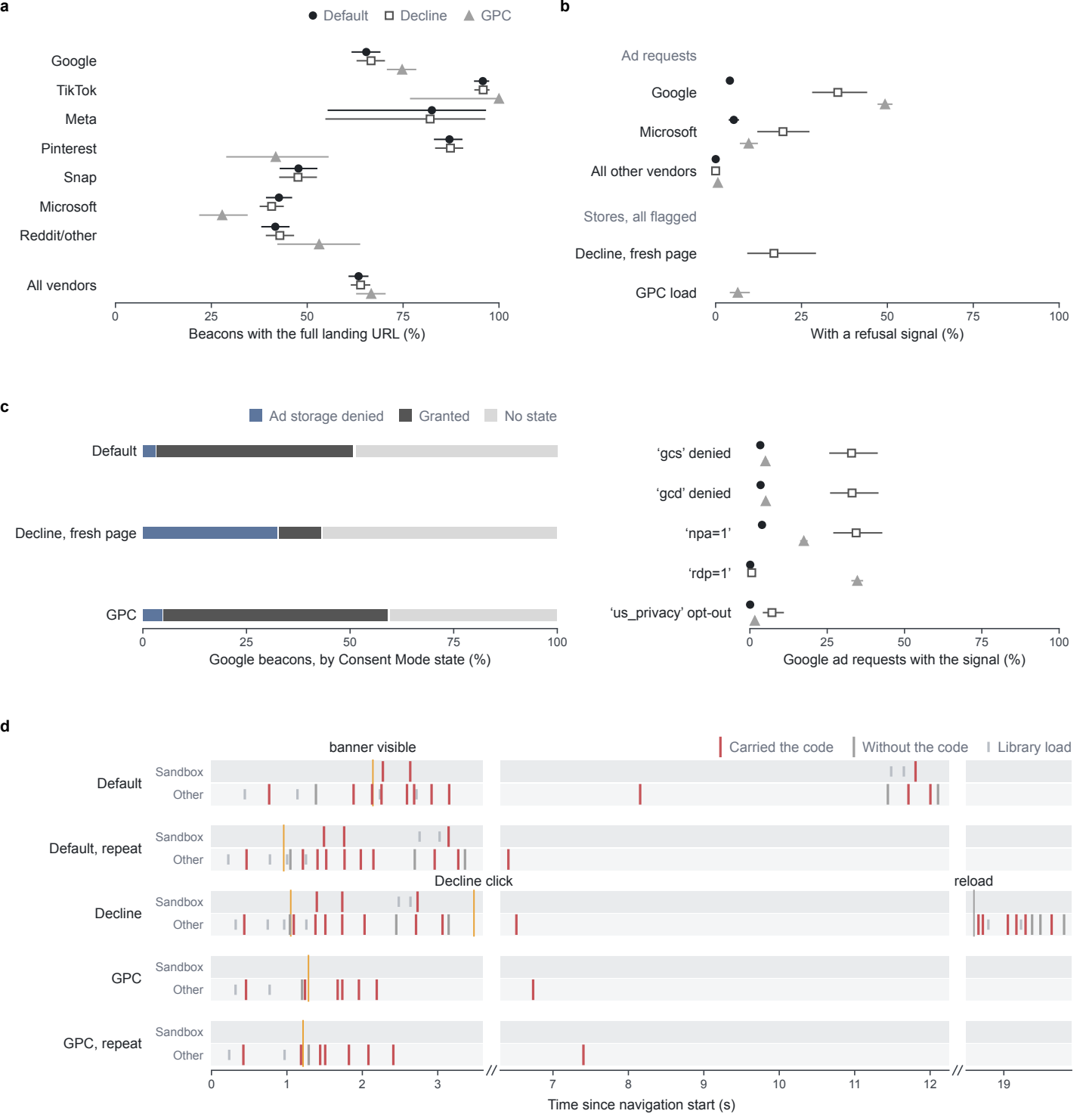

a
Crawler
Yes
No
53
1
not audited
Yes
No
Screenshot audit
b
35
0
1
17
Yes
No
Screenshot audit
c
33
0
0
2
Yes
No
Screenshot audit
d
Cohen's κ
1.00
0.95
0.90
Default vs repeat
GPC vs repeat
GPC vs Brave
Stores
Store × vendor pairs

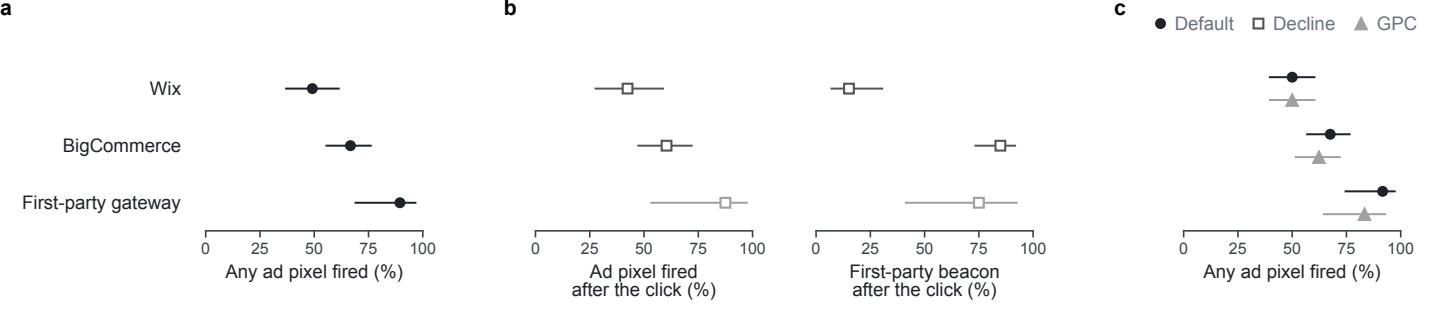

a
Main analysis
Main, bootstrap by consent tool
Crawl segment at 6 browsers
Crawl segment at 10 browsers
Without edge-cached pages
California consent string only
Missed banners added
Consentmo stores as banner stores
Without ethics-flagged stores
Weighted to the Tranco top 200,000
Data beacons only
Beacons with our test code only
Any origin, not only a sandbox
Main analysis
Main, bootstrap interval
Robustness variant
Changed definition
0
25
50
75
100
Sandboxed ad request before any choice (% of stores)
b
Main analysis (fresh page)
Main, bootstrap by consent tool
Including failed forced clicks
Crawl segment at 6 browsers
Crawl segment at 10 browsers
Without any forced click
Without ethics-flagged stores
Weighted to the Tranco top 200,000
Data beacons only
Any time after the click
Fresh page, any sandbox
0
25
50
75
100
Ad request from a new sandbox on the fresh page (% of stores)
c
Main analysis
Main, bootstrap by consent tool
First 10 s of each load
Crawl segment at 6 browsers
Crawl segment at 10 browsers
Without edge-cached pages
Without ethics-flagged stores
Weighted to the Tranco top 200,000
Data beacons only
Beacons with our test code only
Sale-of-data pixels only
Any origin, not only a sandbox
Denominator: all GPC stores
0
25
50
75
100
Sandboxed ad request still sent under GPC (% of stores)


a
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
Any vendor
Banner shown
No banner
0
25
50
75
100
Ad pixel fired (%)
b
banner visible
one poll
Stores, cumulative (%)
100
75
50
25
0
−3
−2
−1
0
+1
+2
+3
First sandboxed request, from banner visible (s)
c
Sandbox
Open pixel
Page script
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
All vendors
0
25
50
75
100
Ad requests (%)
d
Allowed until choice
Denied until consent
Pandectes
OneTrust
Cookiebot
Osano
Shopify's banner
0
25
50
75
100
Any ad pixel fired (%)
e
Banner shown
No banner
Stores with ≥ n requests (%)
100
75
50
25
0
1
10
100
Sandboxed requests per store, n


a
Primary: CMP wrote no 'yes'
All blocklist stores
Vendor's pixel configured
No 'yes' written by the CMP itself
No 'yes' before the pixel started
0
25
50
75
100
Blocked vendor fired (%)
b
Sandbox sent
Yes
No
6
15
2
6
Yes
No
Page script sent
Sandbox
Page script
Not configured
0
25
50
75
100
Sent before any choice (%)
c
Blocklist stores re-checked 21
3 not active
Blocker active in the page 18
10 not neutralised
Blocked vendor's tag neutralised 8
3 sent nothing
Its sandboxed pixel still sent 5
d
Sent
Not sent
Shopify's sandbox
Page scripts
Google
TikTok
Pinterest
Snap
Microsoft
Reddit/other
Pandectes
OneTrust
Pandectes
OneTrust


a
Default
Decline, fresh page
GPC
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
Any vendor
0
25
50
75
100
Stores with a new sandbox request (%)
b
By banner type
Shopify's banner
CMP writes to Shopify
Blocker only or unknown
By Shopify's record
Refusal recorded
'Marketing' allowed
No record
0
25
50
75
100
Stores with a new sandbox request after Decline (%)
c
A refused purpose
Only purposes left allowed
No purpose
Stores still sending (%)
0
5
10
15
20
25
Same page
Unload beacon
Fresh page
GPC load
d
Before a 'no'
After a 'no'
After a 'yes'
consent write
'No', Default
'No', GPC
'Yes', Default
'Yes', GPC
−1
−0.1
0
+0.1
+1
Sandbox start minus the CMP's consent write (s)


a
Default
Decline
GPC
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
All vendors
0
25
50
75
100
Beacons with the full landing URL (%)
b
Ad requests
Google
Microsoft
All other vendors
Stores, all flagged
Decline, fresh page
GPC load
0
25
50
75
100
With a refusal signal (%)
c
Ad storage denied
Granted
No state
Default
Decline, fresh page
GPC
0
25
50
75
100
Google beacons, by Consent Mode state (%)
'gcs' denied
'gcd' denied
'npa=1'
'rdp=1'
'us_privacy' opt-out
0
25
50
75
100
Google ad requests with the signal (%)
d
banner visible
Carried the code
Without the code
Library load
Default
Default, repeat
Decline
GPC
GPC, repeat
Sandbox
Other
Decline click
reload
0
1
2
3
7
8
9
10
11
12
19
Time since navigation start (s)


a
Wix
BigCommerce
First-party gateway
0
25
50
75
100
Any ad pixel fired (%)
b
0
25
50
75
100
Ad pixel fired
after the click (%)
0
25
50
75
100
First-party beacon
after the click (%)
c
Default
Decline
GPC
0
25
50
75
100
Any ad pixel fired (%)